

Changing a Networked Mind: Paradigm Dynamics as Structure Learning over a Hidden World

Jonas Hallgren

David Hyland

Mahault Albarracin*

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Abstract

A scientist is not handed the variables of the world, nor the dependencies among them. The structure of reality is hidden, and a community must *learn* that structure even as it uses it—which means the object a paradigm represents cannot be a fixed menu of hypotheses but a graph that is itself under revision. We take this seriously and build a model of scientific communities as populations of active-inference agents performing *Bayesian structure learning*: each agent carries a precision-weighted network of interdependent commitments, expands it by proposing new nodes and edges, and reduces it by pruning the couplings the data do not support. Two consequences follow without further assumption. First, the exploration that network-epistemology models install by hand—an ϵ -greedy bonus, a switching cost—is here the epistemic drive to expand the model. Second, the network carries *two* fields by one propagation operator: a conservatism field, the cost of revising a commitment given everything that depends on it, and a conviction field, the value attached to a commitment, propagated through the same dependencies. **We implement the model and run it** on a single-agent dark-energy paradigm net and on a population tracking a changing-cosmology world, and report what it does. The two fields come out linearly independent on the dark-energy net—general relativity is the costliest commitment yet only modestly cherished, the cosmological constant the cheapest yet the most cherished—and the supernova degeneracy emerges as exactly the Schur residue the theory predicts. A population locks in only when *conviction and network disconnection act together*: either alone re-tracks a moving world. Adding a forgetting factor breaks the accumulation ratchet so the population re-tracks, and turns lock-in from automatic into *earned*—a conviction threshold, the formal face of motivated persistence. Exploration appears as a Poisson *rate*, and model expansion discovers an unconceived dimension only after it appears. Made to silence its own disconfirming channels, conviction reproduces the predicted evidential lock-in *endogenously*—a γ -crossover, with a deaf-but-honest control confirming the stall is the gating and not absent evidence. We are equally explicit where the model falls short: a conservatism-gated revision rate does *not* recover the belt-first/core-last *staircase*, a clean negative that—because an operator-set Fisher deposit leaves theory identity riding the posterior means—localizes the one missing ingredient to data-dependent structure learning. The Kuhnian phenomenology is recovered as a consequence of bounded structure learning, not posited alongside it.

1 Models of collective belief revision

The question of how a community evaluates, adopts, and discards a system of knowledge has, over the last two decades, migrated from the philosophy of science into computational social science, where the qualitative insights of Kuhn, Lakatos, and Feyerabend have been recast as agent-based and network models one can actually run. It is worth orienting against that body of work before

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adding to it, because the contribution we intend is sharply delimited by what these models already do well and by two things they do not do at all.

The dominant formalism is *network epistemology*: a community of agents, each holding a scalar credence over a small menu of competing hypotheses, repeatedly tests, updates by Bayes, and shares results with network neighbours (Zollman, 2010; O’Connor and Weatherall, 2019). The celebrated results of this program concern the surprising epistemic value of inefficiency. Zollman’s effect—that sparsely connected communities often outperform densely connected ones, because slow information flow shields pockets of the network from early misleading data and so preserves transient diversity—vindicates, in mathematics, the old intuition that a field benefits when some of its members stubbornly resist the consensus. A parallel literature on the *division of cognitive labour* shows that a community hedges its bets best not when its scientists reason as dutiful truth-seekers but when they chase reward, scattering across approaches that pure epistemic rationality would have left unstaffed (Kitcher, 1990; Strevens, 2003). And a more recent strand identifies a whole taxonomy of mechanisms—sparse topology, large populations, strong priors, adaptive learning rules, intermittent ties—that stave off premature convergence (Jonard et al., 2025). Alongside these run the active-inference treatments of collective cognition, in which *epistemic communities* develop shared belief through coupled free-energy minimization (Albarracín et al., 2022), and the recent formalization of belief revision itself as carrying a cost (Hyland and Albarracín, 2025). Decentralized-science efforts, finally, take these coordination questions out of simulation and into the design of real research institutions.

Two features cut across nearly all of this work, and they are exactly the two we wish to change. The first is that *exploration is posited rather than derived*. To keep a community from collapsing onto the first locally attractive theory, these models equip their agents with an exploration term—the ϵ of an ϵ -greedy rule, a temperature, a switching cost, an exogenous prior strength. The diversity that protects the community is bought with a free parameter. The second, and deeper, feature is that *the hypothesis space is fixed*. The agents update over a menu—theory A versus theory B, a handful of arms on a bandit—that the modeller wrote down in advance. The interesting dynamics live in how the topology of communication interacts with movement *within* that menu. What no one studies is a community whose menu is *itself* the thing being learned, revised, and transmitted. The room is nearly empty, and the reason it is empty is that there has been no cheap, principled mechanism for “learning what the model even is” that could be run across a population.

2 Hidden structure and the necessity of structure learning

Return to Kuhn’s deepest observation, which the scalar models gesture at but cannot embody: a paradigm shapes what a scientist can perceive, what she chooses to investigate, and which alternatives she will entertain—three faces, he insists, of a single fact (Kuhn, 1962). The reason a scalar credence over a fixed menu cannot carry this is that a menu has no hidden interior. It can represent which theory is favoured; it cannot represent the dependencies that make one theory *cost* more to abandon than another, nor the way a commitment at the centre of a web of inference quietly determines what counts, at the periphery, as an anomaly worth noticing.

But there is a prior problem, and it is the one we want to make the engine of the paper, because it makes structure learning not a stylistic preference but a necessity. *The structure of reality is hidden*. A scientist confronting a new domain is not given the right variables, and she is certainly not given the dependencies among them. She does not know, in advance, whether two observed regularities share a hidden common cause, are directly and lawfully coupled, or merely coincide. The generative model whose structure she must eventually come to hold is, at the outset, unknown

in its very shape. This poses a question the fixed-menu models never have to face: if you do not know the structure of the world, how do you build a network over it that you can sample from at all?

The answer forces everything that follows. You begin with *some* network—a provisional Bayes net of commitments—and you treat its structure as itself an object of inference, to be expanded when the data demand a commitment you had not represented and reduced when a dependency you had posited earns no evidence. Learning a model from scratch, when the structure is hidden, is *Bayesian model expansion and reduction*. And this immediately discharges the first of our two debts to the prior literature: the exploration term that those models install by hand reappears here with a mechanism behind it. To explore is to *expand*—to entertain a node or an edge not yet in the model—and the drive to do so is the epistemic value of the expansion, the expected information it would yield, which active inference already supplies through expected free energy (Friston et al., 2015; Millidge et al., 2021). We do not add an exploration bonus; we identify exploration with the structural move of proposing a new piece of the world.

Desiderata. The construction that follows is built to satisfy six requirements:

1. that a paradigm be a *structured* object—a network of interdependent commitments with a dense core and a sparse belt in contact with observation;
2. that revising a commitment incur a *carry-over* cost—the aggregate of the re-readings it forces on everything downstream;
3. that *conservatism be derived* from structural position rather than assigned by a dial;
4. that *structure learning be principled*—Bayesian model reduction over a continuous precision space whose corners are the discrete topologies;
5. that *conviction be a second field on the same network*, decoupled from conservatism and written into the model by the same expansion that writes a new belief;
6. that the *population signature* be a graded reorganization, or a stall, rather than a single thresholded flip.

The active-inference machinery enters unmodified throughout. Section 7 reports how each desideratum fares once the model is actually run.

3 Paradigm networks and structure learning

Modeling commitments. We separate what is posited from what is derived.

1. (*Regime*) Couplings are linear-Gaussian: $x = Ax + \zeta$, $\zeta \sim \mathcal{N}(0, D^{-1})$, A a weighted DAG adjacency. Every closed form below—the Schur residue, $\mathbf{T}$, κ , U , and the BMR ratio—holds in this regime; nonlinear couplings would replace $\mathbf{T}$ with a local linearization.
2. (*Acyclicity*) A is a DAG, hence nilpotent, so the Neumann series for $\mathbf{T} = (I - A)^{-1}$ terminates. Cyclic dependence would require the resolvent as a fixed point rather than a finite sum.
3. (*Derived conservatism*) κ is not primitive. It is $\mathbf{T}\mathbf{1}$ —the accumulated record of the expansion-and-reduction process of §3.2, read off at an instant. It is taken as given *within* a round and evolves with the structure *across* rounds (§3.3).

4. (*Exogenous utility*) The intrinsic-utility source u is taken as given. Its endogenous formation—how value accretes onto commitments through planning and credit, progressively aligning u with κ as a programme matures—is outside present scope and is the natural next step.
5. (*Two precision families*) Coupling precisions π_{uv} build the structure, and hence $\mathbb{T}$, κ , and U ; sensory and trust precisions ρ_k, π_s gate evidence, and hence γ . Forgetting (§3.3) acts on the first; gating (§4) on the second.
6. (*Honest agents*) No agent misreports. The only distortions are the value tilt λ and the gating γ , both internal to inference; lock-in is therefore not a dishonesty result.

Let the agent represent a paradigm as a network $G = (V, E, \Pi)$: nodes v carry theoretical commitments x_v , directed edges carry inferential dependencies, and each edge bears a coupling precision π_{uv} measuring how tightly the commitment at v is bound to the one at u . The internal coherence of the paradigm is a precision-weighted graphical model, $p(x \mid G) \propto \prod_{(u \rightarrow v) \in E} \psi_{uv}(x_u, x_v)^{\pi_{uv}}$, so that strongly coupled commitments move together; the dense, high-precision interior is the *core* and the sparse periphery in contact with observation is the *belt*, each belt node reading the world through a sensory channel of precision ρ_k (the same quantity that will weight the social channels of Section 6). Two distinct kinds of change act on this object. The values x_v may be revised under fixed structure, which is ordinary inference; the structure G itself may be revised, which is structure learning. We treat the first kind first, because the cost of revising a value under fixed structure is what the structure-learning moves later weigh.

3.1 The cost of revision: carry-over and conservatism

Revising a commitment is not a local act. Setting $x_v \rightarrow x'_v$ emits precision-weighted messages along the edges incident on v ; each dependent node must re-equilibrate to remain coherent with the new value, and those re-equilibrations emit messages of their own. The cost of the revision is the aggregate of this cascade,

$$C(v) = D_{\text{KL}}[q'(x_v) \parallel q(x_v)] + \sum_{(v \rightarrow w) \in E} \pi_{vw} \mathbb{E}[D_{\text{KL}}[q'(x_w) \parallel q(x_w)]] + \dots, \quad (1)$$

which is the precise, networked content of Kuhnian incommensurability: changing a commitment forces a re-reading of everything that depends on it, and the scalar “cost of changing one’s mind” used by the network-epistemology models is recovered as the degenerate single-node limit of Eq. (1).

This cascade is most legible as linear algebra on the precision matrix. Take a hidden common-cause structure over (x_1, x_2, b) and marginalize the hub:

$$\Pi = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \xrightarrow{\text{marginalize } b} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} = \begin{pmatrix} 1.5 & -0.5 \\ -0.5 & 1.5 \end{pmatrix}. \quad (2)$$

A direct -0.5 coupling is induced between x_1 and x_2 where the joint structure had a zero: marginalizing the hub does not delete its influence but condenses it into a new edge among its neighbours. Carry-over is this Schur-complement residue, and it admits a single closed form once we fix the regime in which Eq. (2) is exact. We work throughout in the linear-Gaussian case, in which the directed structure is a linear structural equation model $x = Ax + \zeta$ with $\zeta \sim \mathcal{N}(0, D^{-1})$ (Koller and Friedman, 2009), A the weighted directed adjacency carrying the coupling precisions (A_{iw} the strength with which i enters its child w). This single matrix A generates both objects the section

uses. It generates the symmetric joint precision $\Pi = (I - A)^\top D (I - A)$, whose Schur complements are the carry-over residues of Eq. (2); and it generates the directed *propagation operator*

$$\mathbf{T} = (I - A)^{-1} = \sum_{k \geq 0} A^k, \quad \mathbf{T}_{iw} = \sum_{\text{paths } i \rightsquigarrow w} \prod \pi = \pi_{i \rightsquigarrow w} \quad (w \in \text{desc}(i)), \quad \mathbf{T}_{ii} = 1, \quad (3)$$

the Neumann series being finite because a DAG makes A nilpotent. The entry $\mathbf{T}_{iw}$ is the total precision-weighted coupling propagated from i to a descendant w along all directed paths. The conservatism of a node is then $\mathbf{T}$ applied to the unit source,

$$\kappa = \mathbf{T} \mathbf{1}, \quad \text{i.e.} \quad \kappa_i = \sum_w \mathbf{T}_{iw} = 1 + \sum_{w \in \text{desc}(i)} \pi_{i \rightsquigarrow w}, \quad (4)$$

the precision-weighted mass of a node’s descendants (the leading 1 being its own unit source). A core node is costly to revise because much depends on it; a belt node is cheap because little does. Conservatism is therefore a derived quantity—the connectivity of the graph read through one operator—rather than a parameter assigned by hand, and it is the first of the two fields the network carries. (We realise $\mathbf{T}$ on the coupling *magnitudes* $|A|$, the positive coupling precisions, so a node’s downstream mass does not cancel when a child opposes a parent; the signed A builds Π and the Schur residue. §7.1 reports both on a worked dark-energy net.)

3.2 Structure learning: reduction and expansion

Structure learning revises G itself, and active inference treats it as the dual of state inference through two complementary operations: pruning structure the data do not support, and proposing structure the data demand (Friston et al., 2017; Smith et al., 2020). The two are asymmetric in cost, and the asymmetry is the formal counterpart of the distinction between normal and revolutionary science.

Reduction is the exact half. Each edge e is parameterized by a prior precision α_e on its parameter, with $\alpha_e \rightarrow \infty$ pinning the parameter to zero (the edge vanishes) and $\alpha_e \rightarrow 0$ freeing it. This is the automatic-relevance-determination parameterization of sparse Bayesian learning (MacKay, 1992; Neal, 1996; Tipping, 2001), in which a prior precision per parameter is used to switch structure on and off; estimating which edges of a Gaussian graphical model to retain is the sparse precision-estimation problem that the graphical lasso (Friedman et al., 2008) solves by L_1 penalization. Bayesian model reduction (Friston and Penny, 2011; Friston et al., 2018) supplies the fully Bayesian counterpart: because a reduced model shares the likelihood with the full model and differs only in its prior, the change in evidence under *any* pruning is available in closed form from the already-inverted full posterior,

$$\frac{p(o \mid \tilde{M})}{p(o \mid M)} = \mathbb{E}_{p(\theta \mid o)} \left[\frac{\tilde{p}(\theta)}{p(\theta)} \right], \quad (5)$$

the posterior expectation of the prior ratio (the Savage–Dickey density ratio in general form). A single inversion thus scores every candidate reduced structure without re-touching the data. The discrete topologies are the vertices of the continuous precision cube spanned by the α_e and the interior is the relaxation into soft, partial edges, so structure learning by reduction is navigation over a precision landscape with an automatic complexity penalty: pruning an edge whose posterior has already collapsed to zero increases the evidence ($\Delta F > 0$), while pruning one the data support decreases it ($\Delta F < 0$). Edge-editing is admissible under Eq. (5) precisely because edges parameterize the prior $p(x \mid G)$ and not the likelihood, whose form the world fixes.

Reduction can only simplify within a model whose dimensions were fixed when the richest candidate was written down; it cannot reach a commitment no one has represented. Expansion supplies the missing dimension, and it is the operation in which exploration, as identified in Section 2, actually consists. Following the model-expansion procedure of Friston et al. (2017), the agent entertains a new commitment by wiring it tentatively to existing nodes with free, uninformative priors ($\alpha_e \rightarrow 0$) and letting reduction prune the proposal back to the couplings the data support (Fig. 1). What to believe about the new node, and how strongly to wire it, are not free hyperparameters but quantities the Bayesian machinery integrates out; the agent chooses only the proposal, and the evidence does the rest. Expansion is not, however, closed-form: a genuinely new node may introduce an observable channel and so enlarge the likelihood’s support, which requires a real inversion, whereas the subsequent pruning is the closed-form half. This cost asymmetry—cheap reduction within a fixed support, expensive expansion of the support—is why a field reorganizes its periphery continually and its centre only rarely.

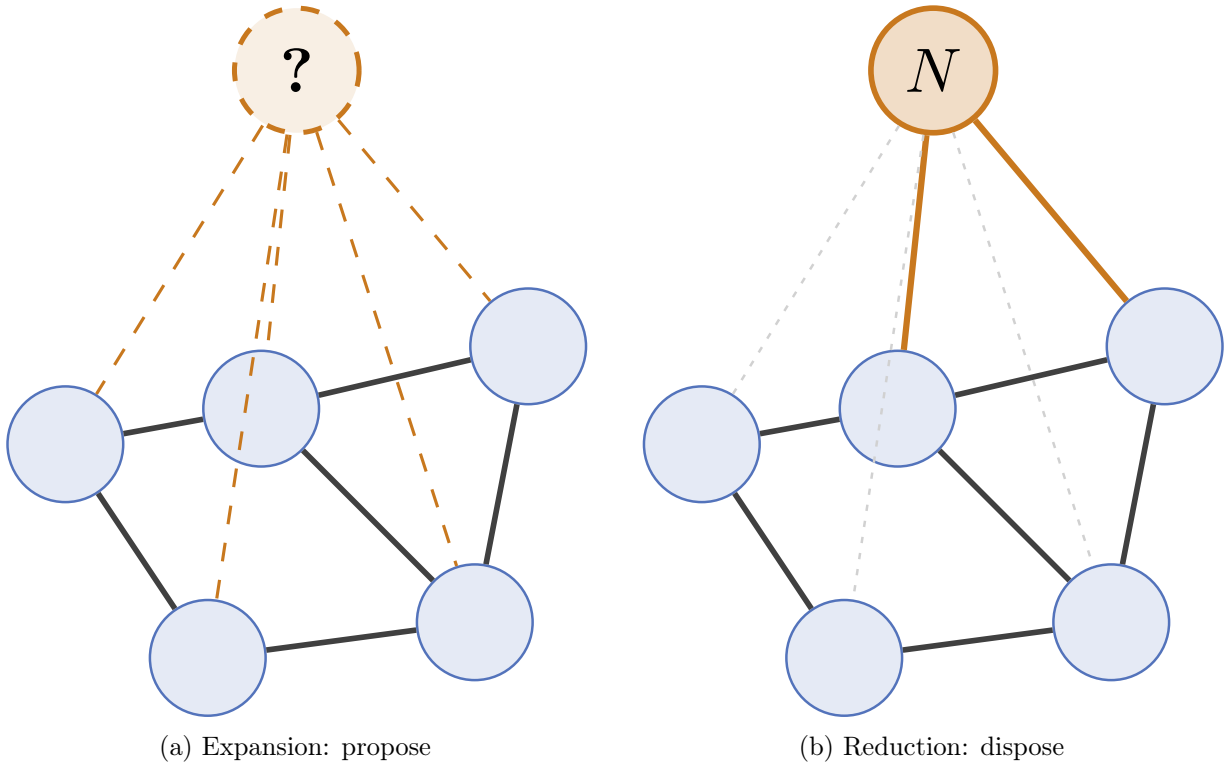


Figure 1: Structure learning as expansion and reduction. a To entertain a commitment not yet in the model, the agent proposes a new node N wired tentatively to every existing commitment by candidate edges with free priors ($\alpha_e \rightarrow 0$, amber, dashed). b Bayesian model reduction prunes the proposal in closed form, keeping only the couplings the data support (solid) and refunding the rest by Occam (faint). Expansion costs a real inversion because the new node may enlarge the likelihood’s support; the subsequent pruning is the cheap closed-form half. Run on a cosmology world in which the truth grows an unconceived dimension, this move discovers that dimension only after it appears (§7.4).

3.3 Temporal structure: accumulation and forgetting

The operator of §3.1 is static. Read once from a fixed set of coupling precisions, $\kappa = \mathbf{T}\mathbf{1}$ can only grow as structure learning adds edges, and nothing lets an unreinforced dependency loosen: a paradigm with no forgetting is a ratchet. This is both unrealistic and—as the population experiments confirm sharply (§7.3)—the degenerate limit in which lock-in is automatic rather than earned.

Active inference carries the missing ingredient and we import it unmodified. In the process-theoretic treatment of learning, the parameters of the generative model are not constants but are carried by conjugate hyperparameters whose concentration accumulates with corroborating evidence; forgetting is a multiplicative *forgetting factor* $\omega \in (0, 1]$ applied to that accumulated concentration between updates (Friston et al., 2017b; Smith et al., 2022). In the discrete case this acts on Dirichlet counts; in the ARD parameterization of §3.2 it acts on the Gamma hyperparameters of the edge precisions α_e , the linear-Gaussian conjugate analogue. Either way an edge precision is the posterior expectation of a parameter whose evidence leaks at rate ω unless refreshed. Concretely, between updates the accumulated information relaxes toward the prior, $\Pi \leftarrow \Pi_0 + \omega(\Pi - \Pi_0)$, so a deposit k steps old carries weight ω^k (an exponential memory of $\sim 1/(1 - \omega)$ steps). The factor is not arbitrary: it is the agent’s prior on the rate of structural drift in the world, the same quantity that volatility estimation makes learnable in the hierarchical Gaussian filter (Mathys et al., 2011)—one forgets at the rate one believes the world’s structure is moving. We sweep ω here and leave its inference to the volatility machinery.

The operator inherits the time index. With A_t rebuilt each round from the decayed-and- reaccumulated precisions, $\mathbf{T}_t = (I - A_t)^{-1}$, and conservatism becomes $\kappa_t = \mathbf{T}_t\mathbf{1}$: a once-central node whose couplings are no longer reinforced relaxes back toward cheapness, and the ratchet is gone. The closed forms of §3–5 are unaffected, because $\mathbf{T}_t$ is fixed *within* a round; the model is a sequence of within-round-stationary problems, and stationarity across rounds was never used.

The decisive choice is what *not* to decay. In its sharpest form one applies the forgetting factor to the coupling precisions alone and leaves the utility source u intact. The asymmetry is not a second free parameter but the mechanism: under $U = \mathbf{T}_t u$ the operator relaxes while the source persists, so a commitment can lose its structural support entirely— κ_t drained, the couplings that justified it forgotten—yet keep its conviction. This is the temporal face of the linear-independence of §4: the static result lets the two fields disagree about *where*, forgetting lets them disagree about *when*, and motivated persistence (a belief wanted-true after its reasons have leaked away) becomes a consequence of the single inequality $\omega_\pi < 1 \leq \omega_u$ rather than a posited bias. In the experiments of §7.3 we use the minimal, symmetric version (one ω on all accumulated information), and find that it already turns lock-in from automatic into a *conviction threshold*; the asymmetric $\omega_\pi < \omega_u$ would make that persistence permanent rather than threshold-gated. Decay applied instead to the *gating* precisions ρ_k would relax γ itself, turning a permanent stall into a race between structural relaxation and conviction reinforcement; we note this as the sharper reading of lock-in.

4 A value field: belief utility and motivated revision

The structure-learning account of Section 3 makes belief change a matter of inferential cost and evidence alone, but precision does not determine which commitments a scientist attends to or propagates. The literature on the division of cognitive labour is explicit that attention is allocated for reasons that are not purely epistemic: scientists distribute themselves across research programmes by pursuing credit and reward under the priority rule, not by maximizing expected information (Kitcher, 1990; Strevens, 2003), and funding priorities, disciplinary prestige, and institutional commitment shape which questions are pursued and which results defended. A result on which a career rests

is held more tightly than its evidence warrants; a programme is sustained because it is fundable. These are not noise on inference but a second organizing field over the same commitments—an attentional or valence field assigning each belief an intrinsic utility independent of its precision. The variational-cost account of belief revision (Hyland and Albarracin, 2025) gives this field its formal handle, treating belief change as a motivated decision that trades the informational cost of revision against the utility of the belief held. It is this value field, and not a refinement of the cost field, that the present section adds.

Active inference already supplies the formal slot for it: the preference matrix $\mathbf{C}$, the log-probabilities of the observations an agent expects and prefers to see. Read through Hyland and Albarracin (2025), this becomes a *belief utility*—a value attached to holding a belief over and above its accuracy, the signature of a boundedly rational agent for whom some conclusions matter more than others. That such a utility should live on a *network* rather than on a single belief is not our invention but an established empirical posture: the Causal Attitude Network model (Dalege et al., 2016) treats a belief system as a network of evaluative reactions in which connectivity is attitude strength, which is exactly the structural reading of conviction we now formalize. The key step is that conviction propagates through the conditional structure by the *same operator* that propagates cost—a commitment one intrinsically wants true lends value to its dependents—so the conviction field is

$$U = \mathsf{T}u, \quad \text{i.e.} \quad U_i = u_i + \sum_{w \in \text{desc}(i)} \pi_{i \rightsquigarrow w} u_w, \quad (6)$$

where u is the vector of intrinsic utilities. The same operator T acts on two source vectors: the unit source $\mathbf{1}$ yields conservatism (Eq. 4), the utility source u yields conviction (Eq. 6). This is the precise content of the two-field claim, and it renders the *decoupling* of Fig. 2 a one-line statement: $\mathsf{T}\mathbf{1}$ and $\mathsf{T}u$ are linearly independent unless $u \propto \mathbf{1}$, that is, unless every commitment is valued in exact proportion to its mere presence. A commitment may be costly to revise yet value-neutral—a deeply embedded technical posit no one has feelings about—or cheap yet cherished, a peripheral belief one is nonetheless attached to. The misalignment of the two colorings in Fig. 2 is exactly this linear independence, and Section 7.1 exhibits it on a real net.

The revision objective and its two cumulants. With conviction in hand we can write the objective the motivated agent actually minimizes, rather than only its solution. When the utility is linear in the belief, belief revision is the minimization of a free energy that trades the fidelity of the honest posterior against the value it can buy,

$$\mathcal{F}[q] = D_{\text{KL}}[q \parallel p(s \mid o)] - \lambda \mathbb{E}_q[U(s)], \quad (7)$$

and the Gibbs variational principle returns its minimizer in closed form, the value-tilted posterior

$$q_\lambda(s) \propto p(s \mid o) e^{\lambda U(s)}. \quad (8)$$

Here λ is the tilt—a single temperature governing how far value is permitted to bend the evidence, an object *entirely distinct* from the structural conservatism κ_i of Eq. (4) despite the family resemblance of the two costs; conservatism is a property of a node’s position in the graph, the tilt a scalar on the whole ledger, and conflating them would wrongly couple the two fields we have just shown to be independent. The normalizer of Eq. (8) is a cumulant-generating function,

$$Z(\lambda) = \log \mathbb{E}_{p(s|o)} \left[e^{\lambda U(s)} \right], \quad (9)$$

and this is what makes the two faces of motivated cognition one object rather than two bolted together. The additive bias that re-centres where the agent lands is the *mean* of the tilted ledger,

$\mathbb{E}_{q_\lambda}[U] = Z'(\lambda)$; the multiplicative bias that sets how strongly any channel can move the agent is its *variance*, the susceptibility $\text{Var}_{q_\lambda}[U] = Z''(\lambda)$. The two biases are the first two cumulants of one tilted ledger. A disconfirming channel held near zero sensory precision ρ_k contributes nothing to that variance, so no tilt along it can move the evidence at all: the agent runs ordinary structure learning, but reads its ledger at a displaced mean and with a curvature its own precision field has flattened along the disconfirming directions.

From gating to a population quantity. It is this gating that becomes, in Section 6, the order parameter γ , and we define it here so that it is built rather than imported. Let the disconfirming sensory channels of a paradigm be those whose evidence would license a core revision, with total precision $\sum_{k \in \text{dis}} \rho_k$. The conviction field can drive a subset of these toward $\rho_k \approx 0$ —curvature flattened where it would hurt most. Define

$$\gamma = \frac{\sum_{k \in \text{dis}} \rho_k \mathbb{I}[\rho_k \text{ gated by the core}]}{\sum_{k \in \text{dis}} \rho_k}, \quad (10)$$

the fraction of the disconfirming sensory precision the core’s conviction field has silenced. Lock-in (Section 6) follows from Eq. (10) through the susceptibility $Z''(\lambda)$: when γ is large, the susceptibility along the disconfirming directions is driven to zero, and a value tilt cannot be displaced by evidence those directions no longer transmit.

5 Coupling of the two fields under model expansion

The two fields are coupled by the expansion operation, and with $\mathbf{T}$ defined the coupling follows directly. When the agent expands its model—proposing a new commitment N wired by candidate conditionals to a set of existing nodes—the joint gains an explicit factor,

$$p(x, x_N \mid G') \propto p(x \mid G) \prod_{(N \leftrightarrow v)} \psi_{Nv}(x_v, x_N)^{\pi_{Nv}}, \quad (11)$$

with the candidate edges entered at free prior precision ($\alpha_e \rightarrow 0$) so that reduction can carve them back. In operator terms, adding N appends a row and a column to A —and therefore to $\mathbf{T}$ —and appends one entry to *each* of the two source vectors, the unit source $\mathbf{1}$ and the utility source u . The two consequences are then immediate. Every node N touches gains the new coupling precision it must now also propagate,

$$\Delta \kappa_v = \pi_{Nv}, \quad (12)$$

while the conviction field gains, at N and along its new edges,

$$\Delta U = u_N + \sum_{w \in \text{desc}(N)} \pi_{N \rightsquigarrow w} u_w. \quad (13)$$

A single expansion thus writes the new commitment into both the cost geometry and the value geometry (Fig. 3). This is the precise content of motivated structure learning: an agent contemplating a node that would, once wired in, lend value to a cherished core—or force the revision of one—registers that prospect in *both* fields, as a carry-over cost it must pay and as a value change it may welcome or resist, and scores the keep-or-prune decision against the value-tilted ledger of Eqs. (7)–(8). The same proposal can therefore be evidence-favoured and value-rejected, or the reverse.

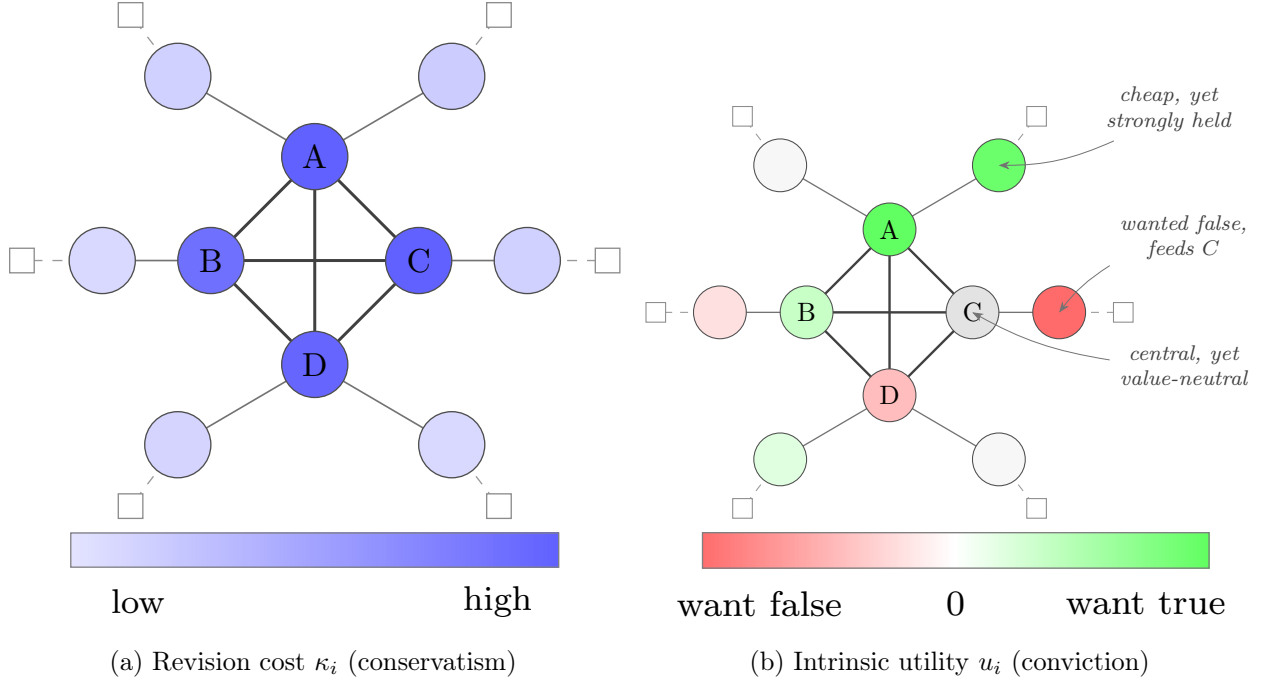
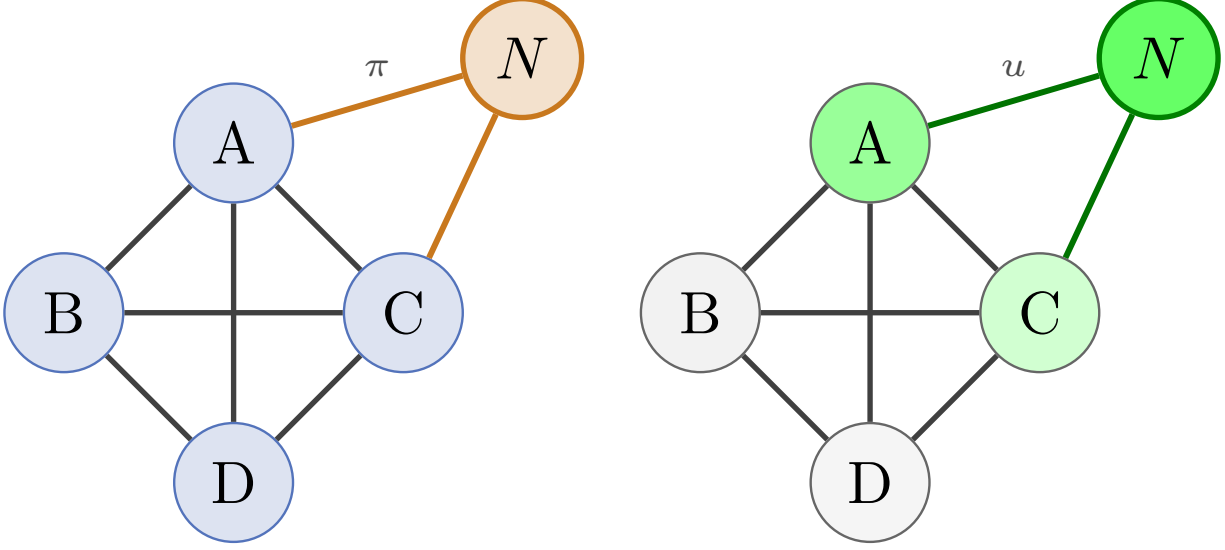


Figure 2: Two fields on one paradigm (schematic). Both panels carry the *same* network—a densely coupled core $\{A, B, C, D\}$ and a sparse belt $\{b_1, \dots, b_6\}$ in contact with observation. a Revision cost is the carry-over of mind-change, $\kappa = \mathbf{T1}$ (Eq. 4): a node’s cost is the precision-weighted mass of its descendants, a clean radial gradient monotone in centrality. b Intrinsic utility is signed—beliefs one wants true (green), wants false (red), or is neutral toward (white)—and the conviction field $U = \mathbf{T}u$ does not align with κ : C is central yet value-neutral; b_2 is cheap yet strongly held; b_4 , which feeds C , is a conclusion the community wants false. The two fields are $\mathbf{T1}$ and $\mathbf{T}u$, the same operator on two sources, and they are independent vectors. Fig. 5 shows this decoupling computed on a real dark-energy net.

6 A population model of collective structure learning

The full model can now be stated. A population of N agents inhabits a shared but hidden world. Each agent carries its own paradigm network over that world—its own structure, its own coupling precisions, its own conviction field $U = \mathbf{T}u$, and a conservatism profile $\kappa = \mathbf{T1}$ read from its connectivity by Eq. (4). Each agent performs structure learning: it samples the world through its belt under expected free energy, prunes by Bayesian model reduction the couplings the data fail to support, and occasionally expands by proposing a new node, writing it into both fields. The agents are coupled. A peer transmits not a scalar opinion but a *candidate structural revision*—a proposed node or edge, together with the precisions it would carry. The receiver treats that transmission as an external source it may trust or distrust, importing the revision weighted by a trust-channel precision π_s ; that is, it performs ordinary active-inference updating on the peer’s proposal exactly as it would on a sensory channel, with π_s playing for the social edge the role ρ_k plays for the senses. The distinction the scalar models could only assert now follows from the representation: a *bubble* is a missing channel, which exposure dissolves, whereas an *echo chamber* is a channel present but held at $\pi_s \approx 0$, which exposure only reinforces (Nguyen, 2018). A third hand acts on this same π_s , external to both the agent and her peers—the *curator*: the recommendation, ranking, and visibility filters of research practice that set, from outside, which revisions reach an agent at all.



(a) enters the belief network (coupling π , $\Delta\kappa_v = \pi_{Nv}$) (b) enters the utility network (propagated value, ΔU)

Figure 3: One expansion, two fields. Adding a commitment N with new conditionals appends a row and a column to the operator $\mathbf{T}$ and an entry to each source vector: into the belief network, where each new edge carries a coupling precision π and so adds to carry-over cost by Eq. (12) (a), and into the utility network, where N 's value propagates along the same new edges by Eq. (13) and tints the commitments it reaches (b). The keep-or-prune decision is scored against the value-tilted ledger, so a proposal can be evidence-favoured yet value-rejected, or the reverse.

The model is built to test whether paradigm learning emerges when a population learns structure together, sharing candidate revisions across a trust network. Because the prediction of interest is organized by conservatism—belt-first, core-last—we resolve the order parameter *by* conservatism rather than collapsing it to a single scalar, which would discard the structural resolution the model is intended to expose. Fix a node alignment between agent a 's network and the rival target structure G^* , and let $\text{align}_i^a(t) \in [0, 1]$ be the fraction of node i 's incident coupling mass whose couplings agree with G^* under that alignment—a structural agreement, not an indicator that a single edge “became” a rival edge. Splitting the nodes by conservatism into a belt and a core, $S_{\text{belt}} = \{i : \kappa_i < \tau\}$ and $S_{\text{core}} = \{i : \kappa_i \geq \tau\}$, define the stratified order parameter

$$m_S(t) = \frac{1}{N} \sum_{a=1}^N \frac{\sum_{i \in S} \mu_i \text{align}_i^a(t)}{\sum_{i \in S} \mu_i}, \quad S \in \{\text{belt}, \text{core}\}, \quad (14)$$

with μ_i node i 's incident coupling mass. A scalar account, with one variable to flip, predicts a single sharp cascade. The structural account predicts a separation between the two curves: $m_{\text{belt}}(t)$ rises first as the cheap periphery reorganizes, and $m_{\text{core}}(t)$ follows only once enough evidence has accumulated to pay the core's carry-over cost (Fig. 4). The gap between the belt and core curves is the staircase. Whether this staircase survives in a runnable linear-Gaussian implementation is one of the questions Section 7 settles—and the honest answer (§7.7) is that it does not, for a reason worth stating precisely.

A third regime is produced by the value field, and its mechanism must be distinguished from that of conservatism. Conservatism alone cannot produce a stall: a high- κ core is merely *slow*—the belt-core gap widens and the delay lengthens—but the disconfirming evidence still arrives, accumulates,

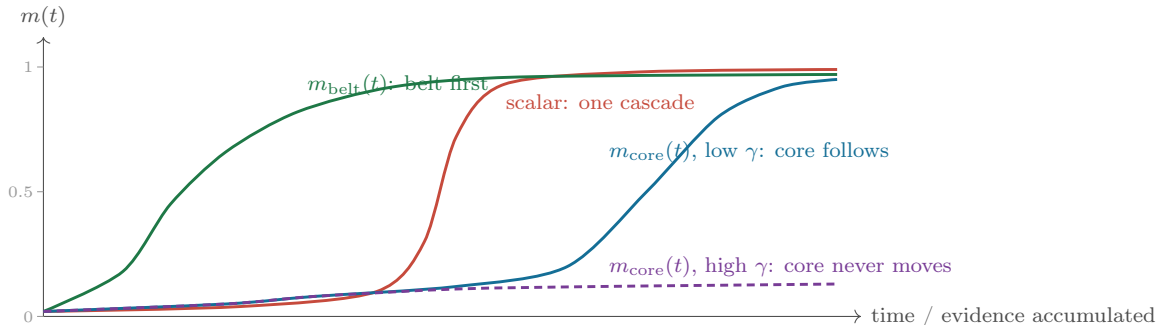


Figure 4: The *predicted* central contrast (schematic). A scalar order parameter (red) has one inflection; the conservatism-stratified order parameter of Eq. (14) is predicted to separate into a belt-first / core-last staircase, with a γ -gated stall at high conviction. Section 7 reports what the implemented model actually does: the population signatures (re-tracking, lock-in, motivated persistence) are recovered, but the staircase by conservatism washes out for the reason given in §7.7.

and the core eventually pays its cost and moves ($m_{\text{core}} \rightarrow \text{high}$, Fig. 4, teal). Permanent stall is a γ -phenomenon. When γ (Eq. 10) is large the channels carrying the rival’s characteristic anomaly are held near zero precision, their susceptibility $Z''(\lambda)$ along the disconfirming directions flattened to nothing, so the cheap belt revisions still happen but the evidence that would license the core revision never arrives, and $m_{\text{core}}(t)$ rises a little and then stalls below unity despite physically present disconfirming data (Fig. 4, violet). Centrality sets how slowly the core moves; conviction sets whether it moves at all. The crossover in γ is a phase boundary, and the stalled core curve—a community that has seen the anomaly and not moved—is the model’s sharpest prediction.

7 Results: running the model

We implemented the model as a linear-Gaussian Bayesian network with the closed forms of §3–5 and ran it in two settings. The first is a single-agent *dark-energy* paradigm net—general relativity (GR) at the root; the density parameters Ω_m and the cosmological constant Λ as dials; the Friedmann expansion, distance modulus $\mu(z)$ and a Type-Ia supernova (SN) reading as a descending chain; light-bending, the perihelion precession, gravitational redshift, CMB and BBN as GR’s other branches; and a modified-gravity (MOND) prediction the community wants false—which makes the two fields and the Schur residue concrete. The second is a population of $N = 150$ agents in three communities tracking a *changing-cosmology* world whose true theory shifts at $t = 60$ and $t = 120$ through dark-matter, modified-gravity and scale-variant regimes. All code is seeded and ships its own null/positive controls. We report what the model does, including—§7.8—where it departs from the schematic prediction of Fig. 4.

7.1 The two fields are realised, and decoupled

Computing $\kappa = \text{T}1$ and $U = \text{T}u$ on the dark-energy net (Fig. 5) makes the two-field claim of §4 concrete. Conservatism is steepest at GR ($\kappa_{\text{GR}} = 10.5$, the maximum) and shallow at the cosmological constant ($\kappa_{\Lambda} = 2.85$, a ratio of 3.7): GR carries every predicted observable below it, Λ only the distance–redshift branch. This is Lakatos’s belt-protects-core falling straight out of the operator—when the 1998 anomaly arrives at the bottom, the cheap move is to turn the dial

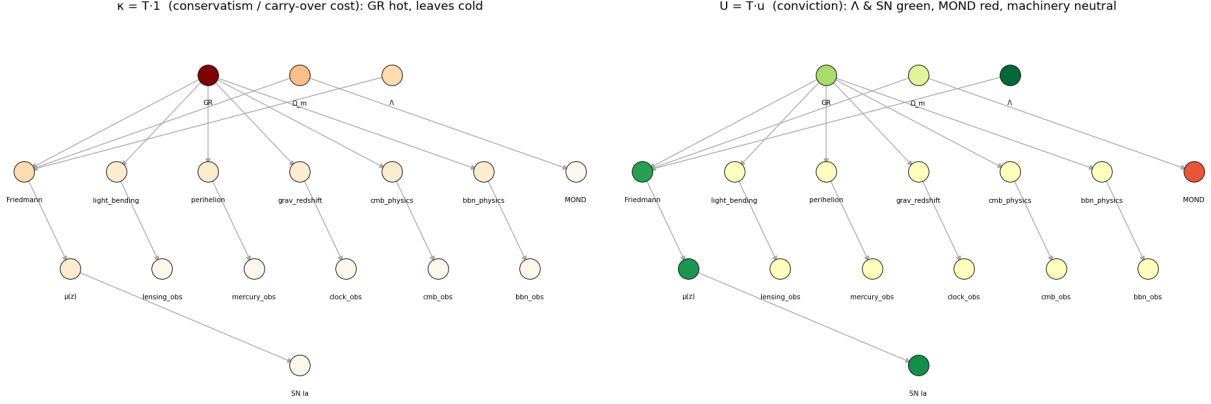


Figure 5: The two fields, computed on the dark-energy paradigm net. *Left*: conservatism $\kappa = T1$ (Eq. 4)—GR is hot ($\kappa = 10.5$), the observation leaves cold; the directed arrows make “everything hangs below GR” visible. *Right*: conviction $U = Tu$ (Eq. 6) on the same graph— Λ and the supernova reading green (cherished), MOND red (wanted false), GR’s other machinery value-neutral. The decoupling is the realisation of Fig. 2: the value order inverts the cost order ($U_\Lambda > U_{GR}$ while $\kappa_\Lambda < \kappa_{GR}$; $r(\kappa, U) = 0.25$).

with the smallest descendant set (Λ), not the ancestor with the largest (GR). The conviction field $U = Tu$, with the community valuing the accelerating-universe supernova reading ($u_{SN} = +1$), the Λ commitment (+0.6) and wanting MOND false (−0.8), *inverts* that order: $U_\Lambda = 1.19$ (the most cherished) exceeds $U_{GR} = 0.45$ while $\kappa_\Lambda < \kappa_{GR}$. The two fields are linearly independent (Pearson $r(\kappa, U) = 0.25$): Λ is *cheap yet cherished*, GR *central yet comparatively value-neutral*, the GR machinery (light-bending, perihelion, clocks, CMB, BBN) central yet value-neutral ($U \approx 0$), and MOND the conclusion the community wants false ($U < 0$)—exactly the three annotated cases of Fig. 2, now read off a real net rather than drawn by hand.

The Schur residue of Eq. (2) is equally concrete. A priori Ω_m and Λ are independent (correlation 0.00, a round cloud). Conditioning on the shared supernova observation—clamping the descendant both dials feed—manufactures an off-diagonal coupling between them where the joint had a zero: the cloud collapses onto the tilted “banana” (Fig. 6), with correlation rising to +0.50. This is the Ω_m – Λ degeneracy direction every dark-energy contour plot draws, here recovered as the Schur complement the carry-over cost is built from—incommensurability and the contour-plot banana the same object, read in two directions.

7.2 Lock-in requires conviction *and* disconnection

In the population, two diversity levers are available: *disconnection* (dropping the cross-community trust density, $\pi_s \rightarrow 0$, so $\lambda_2 \rightarrow 0$) and *conviction* (the per-community value tilt $h \leftarrow h + \lambda U$ toward a home theory). Running all four cells of the 2×2 (Fig. 7) shows the levers fail *differently*, and that a durable, fragmented lock-in needs both. With neither lever, or with disconnection alone, or with conviction alone, all three communities re-track to a single theory by the end of the run (one distinct end-theory). Conviction alone is washed out by precision fusion—the recurring “no-pool” degeneracy in which naive posterior pooling erases heterogeneity. Only with *both* levers do the communities end on three different theories: the entrenched bloc, conviction-tilted to the old incumbent *and*

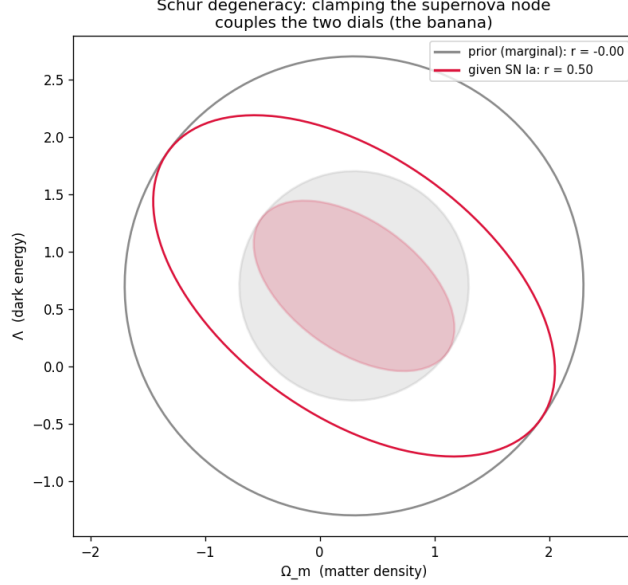


Figure 6: The Schur degeneracy (Eq. 2), computed. The two dials are uncorrelated a priori (grey, $r = 0.00$); conditioning on their shared supernova descendant induces the off-diagonal coupling (crimson, $r = +0.50$)—the Ω_m - Λ “banana”. Marginalizing a shared latent hub and conditioning on a shared observation are the two readings of one complement.

shielded by disconnection from the evidence that would move it, stays on the superseded theory for the whole run, while the others settle on their homes. The residual structural disagreement (Fig. 7, right) stays high (≈ 3.3) exactly in the disconnected cells and collapses to ≈ 0.01 in the connected ones—the fragmented-field signature of lock-in. This sharpens the prediction of §6: disconnection *preserves* structural diversity; conviction *converts* it into a persistent theory-divergence, but only when disconnection protects it from fusion.

7.3 Forgetting breaks the accumulation wall, and makes lock-in earned

The four cells above were run with pure accumulation ($\omega = 1$), and they expose the ratchet of §3.3 in its starkest form: a connected, lever-free population locks onto the *first* theory and never catches the moving truth—early evidence is simply too heavy by the later epochs (Fig. 8, the $\omega = 1$ curve flat at zero after the first shift). Switching on the forgetting factor breaks the wall cleanly: at $\omega = 0.88$ the lever-free population re-tracks each successive theory (fraction on the current true theory rising from 0.00 to 0.92 in the later epochs), because a deposit decays at rate ω and the current epoch can outweigh the stale one. The cost is confidence—steady-state precision falls from $\sim 1.1 \times 10^4$ at $\omega = 1$ to $\sim 5 \times 10^2$ at $\omega = 0.88$ —so there is a sweet spot around $\omega \in [0.88, 0.93]$ that re-tracks while staying confident.

Forgetting also changes the character of conviction, and this is the subtler finding. Without forgetting the value tilt accumulates without bound (it is re-applied every step and never decays), so *any* positive conviction eventually dominates the evidence—lock-in is automatic, a degenerate consequence of the ratchet. With forgetting the tilt saturates at $\lambda U / (1 - \omega)$ and competes with the (also-saturated) per-step evidence, so it holds only above a threshold. Sweeping conviction at $\omega = 0.90$ (Fig. 8, right), the entrenched bloc re-tracks for $\lambda \leq 6$ and stays locked on its home theory for $\lambda \geq 15$. This is exactly the motivated persistence of §3.3—a belief held after its evidence

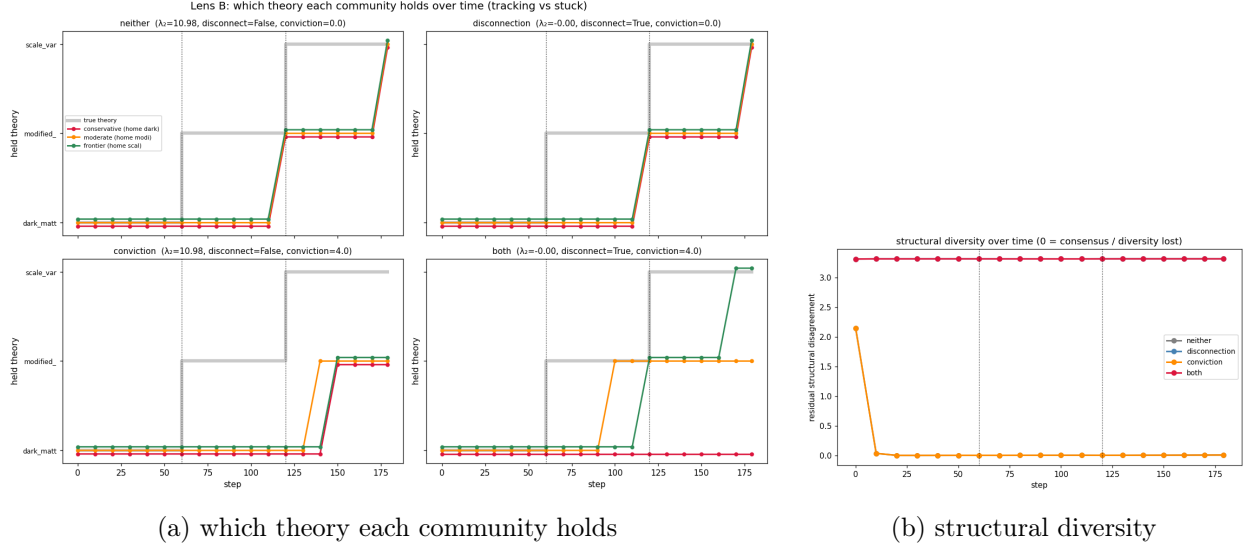


Figure 7: Population tracking of a world that changes theory three times. a Each community’s nearest candidate theory over time; the grey staircase is the truth. Only the *both*-lever cell (disconnected + conviction) ends on three distinct theories with the entrenched bloc stuck on the old one; the other three cells re-track to one theory. b Residual structural disagreement stays high in the disconnected cells (preserved diversity / fragmented field) and collapses to consensus in the connected ones.

has leaked away—turned from a posited bias into a threshold one must clear. (We use a single symmetric ω ; the asymmetric $\omega_\pi < 1 \leq \omega_u$ of §3.3 would push the threshold to zero and make the persistence permanent.)

7.4 Exploration is a rate; expansion discovers the unconceived

The exploration the network-epistemology models post as an ϵ appears here as the *rate* of a Poisson process governing when an agent proposes a structural edit. Letting the discovery of a new node arrive at rate λ , the discovery *time* becomes a waiting time rather than a fixed instant (Fig. 9, left): the mean delay between a cause appearing and the agent finding it tracks $\approx 1/\lambda$, falling from 23 steps at $\lambda = 0.02$ (where 28% of runs never discover the cause within the horizon) to 4 steps at $\lambda = 1.0$. This is the minimal, content-free generator the paper commits to—it fixes the rate and leaves the proposal’s content to the evidence—and it makes “too little exploration” a precise statement: below a rate, the community fails to find what it lacks in time.

What it lacks can be a dimension its menu does not contain. We ran an agent whose six-node cosmology menu has no slot for Λ against a world in which, after $t = 120$, an unconceived dark-energy cause switches on and couples commitments the menu holds independent. The cause cannot be represented as one of the menu’s nodes, so its footprint surfaces as a coherent floor in the agent’s prediction errors; when that floor clears the expansion trigger the agent *wakes a new node*—it grows the dimension (Fig. 9, right). The wake fires at $t = 133$, thirteen steps after the cause appears and never before; the drive-marginal edge- F_1 jumps from 0 to 1; below a coupling strength of ≈ 0.8 the cause is too weak to be discovered, and a null world with no cause never wakes. This is Stanford’s problem of unconceived alternatives made operational: the dimension no one has drawn is reached only by expansion, and only once the world makes its absence felt.

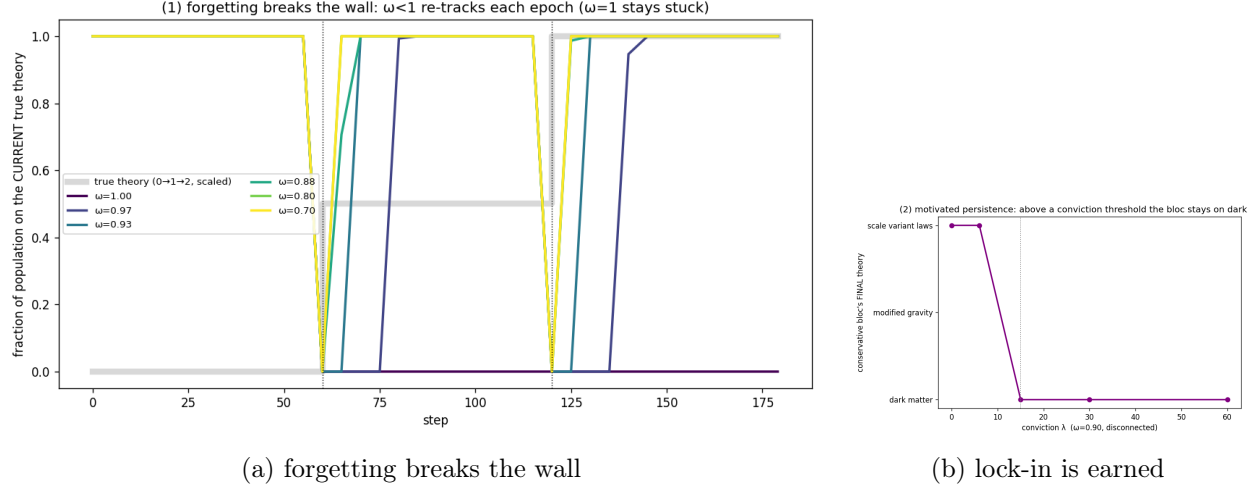


Figure 8: Forgetting (precision relaxation), in the lever-free population. a With $\omega = 1$ the population is stuck on the first theory; with $\omega < 1$ it re-tracks each epoch (fraction on the current true theory), faster forgetting recovering sooner. b At $\omega = 0.90$ with disconnection, the committed bloc re-tracks for small conviction but stays on its home theory above a conviction threshold ($\lambda \gtrsim 15$): lock-in is now *earned*, not automatic—motivated persistence.

7.5 A coarse model resolves a complex world only slowly

A last experiment isolates the cost of a coarse Bayes net against a world richer than it. The true world is a complex twelve-parameter Gaussian with four latent factors (correlated, individually ambiguous directions); a coarse agent senses only res random aggregate combinations per step. Coarser is slower (Fig. 10, left): the time to resolve the world state to within a fixed tolerance grows from 58 steps at full resolution to 221 steps at $\text{res} = 1$, and below a resolution the agent does not finish within the horizon. Adding forgetting floors the resolution outright (Fig. 10, right): a net that both under-senses and forgets plateaus at a residual error that rises as ω falls ($0.08 \rightarrow 0.28$), and at $\text{res} = 3$, $\omega \leq 0.93$ never resolves the world at all. A complex world has no clear answer for a bounded, forgetful, under-sensing learner—the honest limit the hidden-structure premise of §2 implies.

7.6 Endogenous γ : lock-in derived from conviction, not imposed

The lock-in of §7.2 was driven by exogenous levers. The paper’s sharper claim (Eqs. (10)–(9)) is that conviction *itself* silences the channels that would refute it: a community that values the incumbent drives the disconfirming sensory precisions ρ_k toward zero, the susceptibility $Z''(\lambda)$ collapses along those directions, and the core stalls though the evidence is physically present. We make this endogenous. Each round the observation weight on the disconfirming rows is recomputed as a saturating function of the conviction field projected onto them, $w_k = \exp(-g |U \cdot H_k|)$, so $\gamma = \overline{1 - w_k}$ rises with the conviction-to-censorship coupling g ; no channel is gated by hand. On the phlogiston paradigm (whose disconfirming rows genuinely carry the refuting mass-balance signal) the result is the γ -crossover the theory predicts (Fig. 11): as g rises, endogenous γ climbs $0 \rightarrow 1$ and the structural revolution collapses from a revolted fraction of 1.0 to 0.0. The decisive control is *deaf-but-honest*: at the *same* conviction with the gate switched off, the channels stay open and the community revolts (1.0)—so the stall is the conviction-driven gating, not absent or unreachable

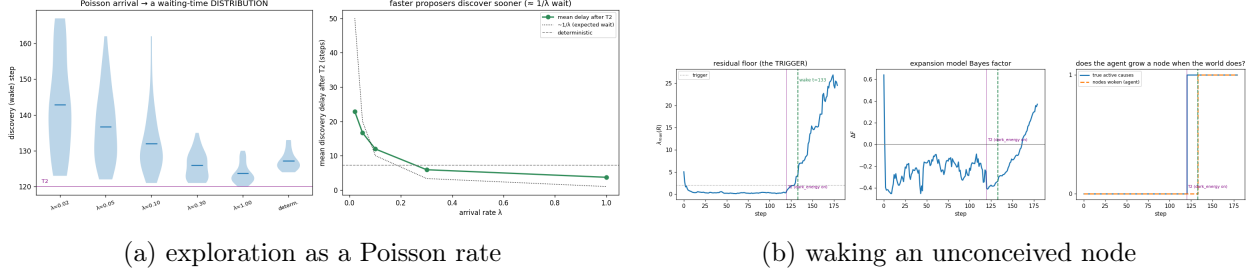


Figure 9: Exploration and expansion. a Proposal arrivals as a Poisson process: the discovery delay after a cause appears is a waiting time of mean $\approx 1/\lambda$; at low λ some runs never discover the cause within the horizon. b An agent whose menu lacks Λ wakes a new node *after* the unconceived cause appears at $t = 120$ (here $t = 133$), never before; the residual floor (left), the expansion Bayes factor (middle) and the node count (right) move only once the world makes the missing dimension felt.

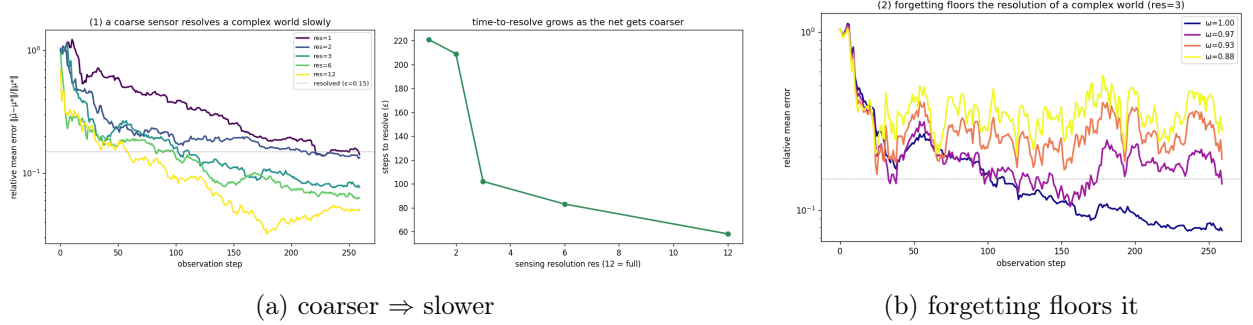


Figure 10: A coarse net on a complex world. a Relative error in the reconstructed world state over time, by sensing resolution; time-to-resolve grows sharply as the net coarsens. b With a fixed coarse sensor, forgetting plateaus the error at a floor that rises as ω falls—the world is never fully resolved.

evidence. This discharges the caveat that lock-in was merely lever-imposed: it is derived from γ via $Z''(\lambda) \rightarrow 0$, with honest agents and reachable truth, exactly the paper’s sharpest prediction.

7.7 Does a conservatism-gated revision rate recover the staircase? No—and that localizes the gap

The other prediction of Fig. 4—a belt-first, core-last staircase in the conservatism-stratified order parameter—does not appear in the base dynamics because κ is a static read-out that never enters the update. The natural fix is to let κ gate the *rate* of revision: a per-node deposit gate $g_i = 1/(1 + \beta\kappa_i)$, PD-preserving, so a high- κ core node integrates evidence slower than a cheap belt node. We tested it on phlogiston, and it does *not* recover the staircase (Fig. 12). The belt $\rightarrow$ core lag stays negative (-55 steps at $\beta = 0$, -45 at large β): the high- κ nodes reorganize *first*, not last, and the κ -gradient gate is indistinguishable from a gradient-free uniform slowdown (-45) and a shuffled- κ gate (-45). Two reasons, and they unify this section’s two negatives into one. First, a deposit gate scales both numerator and denominator of the mean $\mu = \Pi^{-1}h$, so it barely throttles mean-tracking—and mean-tracking is where theory identity lives. Second, the nodes that actually reorganize are exactly the high- κ mass-law commitments the relational operator refutes directly, while the low- κ belt are agreement nodes with nothing to reorganize, so the conservatism stratification is confounded with move-versus-don’t-move. Both reasons are one root cause: with an operator-set Fisher deposit

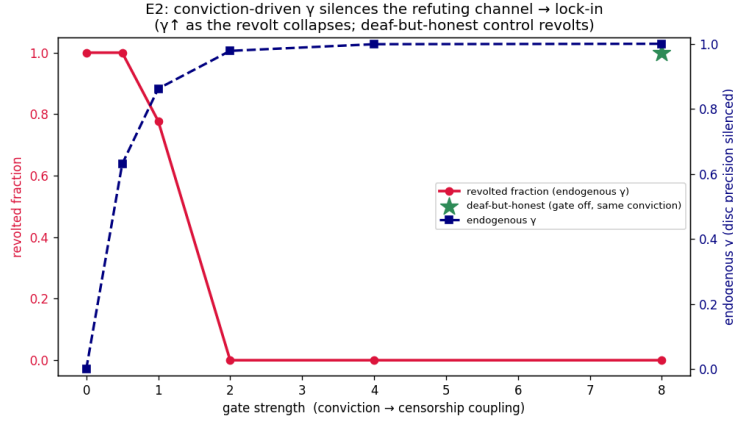


Figure 11: Endogenous γ (phlogiston). As the conviction-to-censorship coupling g rises, the conviction field silences its own disconfirming channels: γ (navy) climbs to 1 and the revolted fraction (crimson) collapses through a sharp crossover—an evidential lock-in *derived* from conviction. The deaf-but-honest control (green star: same conviction, gate off) revolts, proving the stall is the gating and not absent evidence.

$J = H^\top H / \sigma^2$ —set by the observation operator, not the data—the structure does not move, so theory identity rides the posterior *means* and neither a rate gate on those means nor the stratified couplings can express the staircase. The conservatism gradient κ is real and realised statically (§7.1); the negative tells us precisely what is missing to make it temporal: a *data-dependent* observation model in which the couplings themselves are learned (a structure-matched or nonlinear operator). That is the central modelling target the result identifies.

7.8 What remains stylized

Two of the section’s caveats are now discharged or sharpened: lock-in is *derived* from γ (§7.6), and the staircase’s absence is *localized* to the operator-set Fisher deposit (§7.7). One general caveat stands.

It is a stylized model. The dark-energy net is illustrative, its weights chosen to expose the geometry rather than to fit data; the three dials (ω , λ , sensing resolution) are swept, not inferred; and a community that “tracks” the eventually-true theory is committed to it, not thereby correct. Nothing here is evidence about cosmology—it is a model of how a structured, bounded, valued mind changes.

8 Relation to prior work

The differentiator from the network-epistemology and reinforcement-learning literatures is not the tool but the explanatory claim, and it is exactly the two debts of Section 1 now discharged. Where those models *posit* an exploration bonus and a switching cost, we *derive* exploration as the epistemic drive to expand the model (realised as a Poisson rate in §7.4) and the cost of switching as the carry-over of a structural edit, $\kappa = \mathbf{T}\mathbf{1}$ (realised in §7.1). Where those models fix the hypothesis space, we make it the object of inference—and §7.4 shows the menu itself growing a node. Much of the accumulated literature recovers as a special case of the one mechanism. The Zollman speed–accuracy trade-off is the statement of how the algebraic connectivity of the trust graph governs the

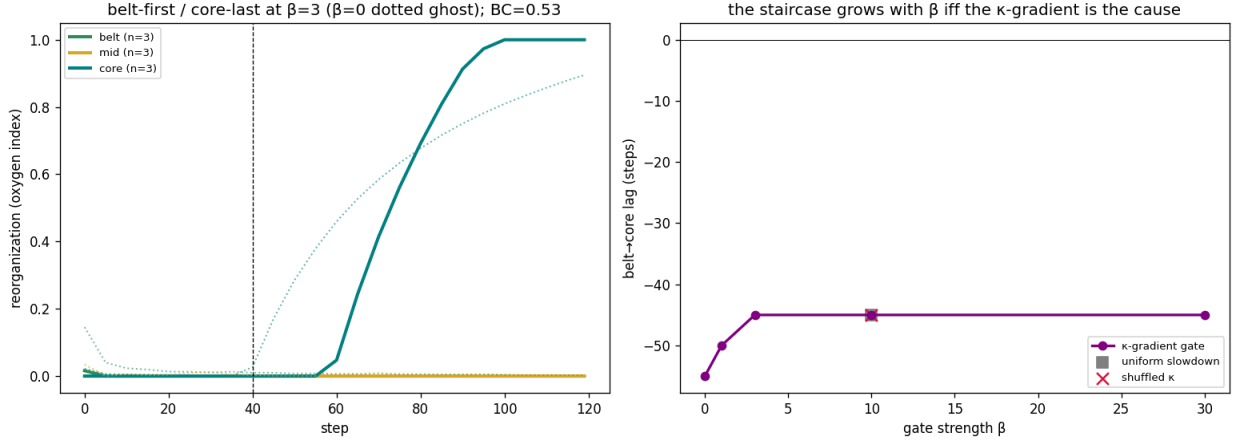


Figure 12: The staircase is *not* recovered by a conservatism-gated revision rate (phlogiston). *Left*: reorganization by κ -shell at the best β ($\beta = 0$ baseline dotted)—the high- κ “core” moves first and the low- κ “belt” (agreement nodes) never reorganizes, the reverse of the prediction. *Right*: the belt→core lag stays negative and the κ -gradient gate (purple) is matched by the gradient-free uniform (grey square) and shuffled- κ (red cross) nulls—the gradient adds nothing. The negative localizes the missing ingredient to data-dependent structure learning.

transient—denser graphs reorganize faster but lose the transient structural diversity that protects against locking onto the worse paradigm, exactly the disconnection lever of §7.2; swept directly on the cosmology task, raising λ_2 drives the population’s residual structural diversity from 3.3 to 0.2, and sharing whole posteriors (conclusions) rather than raw Fisher (evidence) herds it the same way ($3.3 \rightarrow 0.5$)—the content-versus-conduit reading of why sharing conclusions propagates bias. The paradoxical finding that confirmation bias can improve group learning is the same fact read at the agent level, high- κ_i agents preserving diversity, with evidential lock-in as its pathological limit. The bubble-versus-chamber distinction (Nguyen, 2018) is the formal separation between the sparsity of trust channels and the precision they carry. And Thagard’s connectionist model of explanatory coherence (Thagard, 1989) is recognizable as the precision-weighted network $p(x \mid G)$ with the gains flattened and the dynamics frozen—a paradigm net without the active sampling that makes our account move.

9 Discussion

The scalar treatment with which the field began is the degenerate single-node limit recovered when the paradigm has no internal structure, hence no channels to turn down. Everything distinctive follows from taking the hidden structure seriously: a conservatism gradient derived rather than assigned (one operator on the unit source), an exploration drive that is the value of expanding the model rather than a tuned bonus, a cost of change that is the literal re-reading of dependent commitments, a second field of conviction that is the same operator on a different source and decouples cleanly from conservatism on the same graph, and a population signature—graded reorganization, or its stall into lock-in—distinct from a damped oscillation to consensus. Running the model bears the qualitative predictions out: the two fields come apart on a real net (Fig. 5), lock-in is a genuine and structured phenomenon requiring conviction and disconnection together (Fig. 7), forgetting both restores re-tracking and converts lock-in from automatic into earned (Fig. 8), and expansion

discovers a dimension the menu lacked (Fig. 9). It also tells us precisely where the idealisation must be repaired—the operator-set Fisher deposit that flattens the staircase (§7.7)—which is more useful than a clean confirmation would have been.

The Kuhnian reading is now a corollary. Normal science is exploitation: reduction and value-fitting within a fixed cube of hypotheses. Revolution is exploration: the costly expansion that adds a dimension the cube did not contain. Incommensurability is the off-diagonal residue of a Schur complement, the carry-over edge one paradigm’s structure deposits when read through another’s—the very banana of Fig. 6. And lock-in is the collective amputation of a correction channel.

This also locates what remains genuinely uncertain in a way that resists being made Bayesian, and it is the point on which the paper opened. Structure learning navigates beautifully within the dimensions a model posits, but it is blind to the dimension no one has proposed—the node not yet drawn even as a dashed candidate. That frontier is the Knightian residue, and it is precisely Stanford’s problem of unconceived alternatives (Stanford, 2006): not uncertainty over which known structure is right, which the precision cube renders measurable, but uncertainty over whether the space of known structures contains the truth at all. A community doing only reduction is in permanent normal science; the unmeasurable possibility that the whole cube is misspecified is what expansion, and only expansion, can resolve. One cannot place a posterior over the unconceived; running an expansion process that keeps proposing nodes—at some Poisson rate—is what a healthy epistemic system does instead of pretending it can, and §7.4 shows both that it works and that too low a rate leaves the dimension undiscovered. The hidden structure of the world, which forced us into structure learning on the first page, is in the end the thing no amount of structure learning can guarantee to have caught—and it is on that frontier that the curator’s external hand on π_s becomes most consequential, able to push a population across the lock-in boundary from the outside by ensuring that the revisions which would expand the cube are never sampled.

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